



Mobile phone ownership, social media use, and substance use at ages 11–13 in the ABCD study

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ABSTRACT

Introduction: There is ongoing concern about the impact of increasing use of social media and digital devices on unhealthy behaviors such as substance use in youth. Mobile phone and social media use have been associated with substance use in adolescent and young adult samples, but few studies have evaluated these relationships in younger samples.

Methods: This secondary analysis of data drawn from the ABCD Study examined associations between youth-reported mobile phone ownership and social media use at age 11–12 and use of alcohol, nicotine/tobacco, and cannabis over the next 18 months.

Results: Longitudinal logistic regression was used to test the hypothesis that phone ownership and social media use at age 11–12 would predict substance use over time. Phone ownership was associated with greater odds of alcohol and nicotine/tobacco use, and social media use was associated with greater odds of using nicotine/tobacco and cannabis.

Conclusions: These findings suggest that pre-teen youth who own mobile phones and those who use social media may be at greater risk for substance use. Further research is needed to specify mechanisms by which this association occurs and thus inform prevention and intervention efforts.

1. Introduction

Adolescent access to mobile devices and social media platforms has increased over time (Kwan et al., 2020), leading to profound changes in important aspects of development, including how teenagers access information, how they engage in social relationships, and their cognitive development (Nasi & Koivusilta, 2013; Senekal et al., 2023; Sparrow et al., 2011). Combined with evidence of concurrent increases in mental health concerns in younger populations, this has led to debate about the impact of social media and digital devices (Twenge et al., 2018). Overall, research suggests that social media and digital activity during this key developmental period confer both risks and benefits (Choukas-Bradley et al., 2023; Valkenburg et al., 2022).

Social learning theory posits that youth behavior is partially shaped by the behavior and approval of others, via modeling and social reinforcement (Bandura, 1977). An extensive literature has documented that youth substance use is influenced by actual and perceived peer use (Henneberger et al., 2020; Simons-Morton & Chen, 2006; Wang et al., 2022) and by exposure to use behavior and positive portrayals of use in

media such as movies and television (Hanewinkel et al., 2014; Leonardi-Bee et al., 2016). The rise of handheld digital devices and social media has altered the social learning landscape. Research suggests that online posts depicting substance use primarily present a positive image of use (Rutherford et al., 2023). Greater access to digital platforms may increase exposure to positive messages about substance use from celebrities, notable digital “influencers”, marketing campaigns, and others to whom youth are not connected in their non-digital lives, as well as from their friends and peer group (Jackson et al., 2018; Stevens et al., 2020). Exposure to digital devices and platforms may act as a “super peer”, facilitating exposure to positive portrayals of substance use and social reinforcement of use from a wider social network including not only “real world” peers but celebrities and social media personalities (Elmore et al., 2017; Vogel et al., 2024).

Exposure to digital messages about substance use has been shown to impact substance-related behaviors and cognitions (Purba et al., 2023). For example, exposure to alcohol consumption on social media increases the perceived prevalence of peer use (Litt et al., 2023), which is itself a predictor of adolescent substance use (Simons-Morton & Chen, 2006). A

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recent meta-analysis demonstrated strong associations between both exposure to and posting of online alcohol-related content and subsequent alcohol use (Cheng et al., 2024). Similarly, recent studies have found that more frequent social media use (Ranker et al., 2024) and more frequent social media exposure to nicotine/tobacco content (Vogel et al., 2024) both prospectively predict nicotine/tobacco initiation among youth without prior nicotine/tobacco use. A meta-analysis of 29 nicotine/tobacco studies demonstrated that social media exposure to nicotine/tobacco content predicted a doubling of the likelihood of use of multiple nicotine/tobacco products, and increased susceptibility to future use among those without any use history (Donaldson et al., 2022). Finally, exposure to pro-cannabis social media has been associated with stronger intent to use cannabis in the future, stronger beliefs that use is normative among peers, and with use frequency (Willoughby et al., 2024). Youth who use social media more frequently are more likely to subsequently initiate cannabis use (Lee et al., 2023).

A growing literature suggests that youth who use digital devices and social media may be more frequently exposed to positive messages about alcohol, nicotine/tobacco, and cannabis, and that these exposures increase susceptibility to and use of these substances (Donaldson et al., 2022; Rutherford et al., 2023). However, previous studies have typically focused on individual substances, and on older adolescents and/or young adults (Purba et al., 2023). Less is known about these relationships in early adolescence, or about the relative strengths of associations across alcohol, nicotine/tobacco, and cannabis. Additionally, we have limited knowledge about the extent to which substance use is associated with use of internet-connected devices generally versus social media use specifically. Both would seem to potentially increase exposure to pro-substance messages and thus risk of substance use. However, social media use likely leads to exposure to influencers and industry marketing that may be particularly impactful (Laestadius et al., 2019; Vassey et al., 2023) compared with messages received via use of a mobile phone for communication with friends and family for youth who may own phones but not use social media. A better understanding of the potential influence of mobile devices and social media use on youth substance use is critical, given that substance use disorders during adolescence are associated with risk for negative long-term outcomes, including substance use disorders and poorer cognitive functioning and physical health (Hu et al., 2020; McCabe et al., 2022; Wilson et al., 2021).

The current study was a secondary analysis of data from the Adolescent Brain Cognitive Development (ABCD) Study, a nationally-representative, multi-site longitudinal cohort study that enrolled children initially aged 9–10 at 21 sites beginning in 2016. We tested whether mobile phone ownership and social media use at approximately age 11–12 (i.e., at the Year 2 annual assessment) prospectively predicted use of alcohol, nicotine/tobacco, and cannabis over the next 18 months (i.e., approximately ages 12–14). We hypothesized that youth who owned phones and youth who reported social media use at age 11–12 would each be more likely to report any use of alcohol, nicotine/tobacco products, and cannabis over time.

2. Methods

Sample and Protocol. The ABCD Study is a longitudinal cohort study with a sample of youth ($n = 11,880$) who were aged 9–10 years at enrollment. The overall ABCD Study utilized an epidemiologically-informed plan to recruit and enroll participants at 21 sites across the United States (US) to ensure a diverse sample that is broadly representative of the US demography (Garavan et al., 2018). ABCD participants complete annual comprehensive assessments in-person or via video-conferencing, and briefer mid-year assessments via telephone 6 months after each in-person annual follow-up visit. The present study began at the age 11–12 annual visit (i.e., the Year 2 annual assessment in the larger ABCD Study), when youth participants were first asked about mobile phone ownership and social media use. The present study and also included the next three assessment points (i.e., Year 2 mid-year

assessment, Year 3 annual assessment, and Year 3 mid-year assessment); participants were 13–14 years old at the last assessment point. Data described in this manuscript were collected between September 2018 and January 2022. The sample for the present study included all participants from the overall ABCD study who provided substance use data at one or more of these four assessments ($n = 10,907$). The Institutional Review Board at the University of California, San Diego approved all aspects of this study for the ABCD consortium. Both youth assent and parental consent were obtained prior to study participation.

Assessment of Predictor Variables. At the age 11–12 annual assessment, participants were asked two binary questions that were used as predictor variables: 1) Do you have your own mobile phone? 2) Do you have at least one social media account? (Bagot et al., 2022). For the purposes of the present study, these binary variables were used as measures of mobile phone ownership and social media use, respectively. Both were coded as 0 = no phone/no social media use, 1 = own a phone / have social media account(s).

Demographic Variables. Date of birth, sex assigned at birth, racial/ethnic background, and household income were measured at the overall ABCD study baseline. For the purposes of the present study, age was calculated as age in years at the time of the age 11–12 assessment. Sex assigned at birth was coded as 0 = male, 1 = female. Racial/ethnic background was coded as a categorical variable with five levels (Black or African American, Hispanic or Latino, Asian, non-Hispanic White, Other). Household income was coded as an ordinal variable from 1 to 10, with higher values reflecting higher reported household income.

Substance Use Measures. At the age 11–12 and age 12–13 annual assessments, the Timeline Followback (Robinson et al., 2014) was administered to obtain interviewer-assisted youth self-report of alcohol, nicotine/tobacco, and cannabis use, including days of use and number of use episodes per day. At the two mid-year assessments, a brief substance use questionnaire asked whether participants had used each substance in the past 6 months (i.e. since the prior assessment) (Lisdahl et al., 2021; Sullivan et al., 2022). These measures were combined to create time-varying binary variables indicating whether or not participants reported any use of alcohol, nicotine/tobacco, and cannabis since last assessment.

Analytic Plan. To estimate associations between predictors and substance use outcomes over time, we utilized a longitudinal logistic regression model for correlated dichotomous outcomes, estimated using the generalized estimating equations (GEE) method (Liang & Zeger, 1986), implemented via the *xtgee* module in Stata 17.0 using a binomial distribution with a logit link. An autoregressive covariance structure was employed to account for within-participant correlation between repeated outcomes over time. Separate models were fit for each unique combination of predictor (phone ownership, social media use) and outcome (alcohol, nicotine/tobacco, cannabis use), yielding a total of six models. In each model, the outcome variable was time-varying, including all four study timepoints, and predictor variables and covariates were time-invariant and were measured at age 11–12, the analytic baseline of the current study. Each model included all observed data in an “intent-to-treat” approach, with no imputation of missing data.

Each model initially included terms for the predictor, linear time, and the interaction between the two. The time variable was coded from 0 (analytic baseline, participant age 11–12) to 3 to facilitate interpretation. If the interaction term was non-significant, it was not retained. Significant interaction terms were interpreted via simple effects tests that evaluated change over time separately for each level of the predictor. The literature demonstrates that youth substance use rates, social media use, and smartphone access may all differ by age, sex, race, and socioeconomic status (Alexander et al., 2024; Donaldson et al., 2023; Pew Research Center, 2021a, 2021b, 2024; Sullivan et al., 2022). Given their potential associations with both the predictor and outcome variables, each model initially included age, sex assigned at birth, racial/ethnic background, and household income as covariates to reduce the possibility of confounding. Previous racial/ethnic background was

coded as a categorical variable, with non-Hispanic white as the reference group. Alpha was set at 0.05 for all hypothesis tests.

3. Results

Sample Characteristics. Demographic and clinical characteristics are shown in Table 1. At age 11–12, the majority of participants owned mobile phones, and about half reported use of social media. Bivariate chi square tests indicated that participants who owned mobile phones at age 11–12 were significantly more likely to report use of alcohol, nicotine/tobacco, and cannabis, as were participants who were social media users at that age ($p < 0.01$). A separate chi square test of proportions indicated overlap between the two predictor variables [$\chi^2(1) = 1531.05$, $p < 0.001$]. More specifically, at age 11–12 social media use was endorsed by 66.7 % of participants who owned a mobile phone, and 25.7 % of participants who did not. Conversely, 86.6 % of social media users also endorsed phone ownership, compared with 52.8 % of those who reported no social media use.

Mobile Phone Ownership. The prevalence of use of alcohol, nicotine/tobacco, and cannabis over time by phone ownership at age 11–12 is shown in Fig. 1. First, we examined the association between phone ownership and alcohol use (Table 2). The phone*time interaction was not significant and was not retained. There was a main effect of phone ownership ($b = 0.29$ [95 % confidence interval 0.16, 0.41], Odds Ratio [OR] = 1.34 [95 % confidence interval 1.18, 1.51], $p < 0.001$), such that participants who owned phones at age 11–12 were about 34 % more likely to report any alcohol use across the study period.

Next, we tested the association between age 11–12 mobile phone ownership and nicotine/tobacco use over time. In the initial model, the interaction between phone ownership and time was not significant and was not retained. The final model is shown in Supplementary Table 1. This model indicated a non-significant main effect of time ($b = 0.13$ [-0.01, 0.27], OR = 1.14 [0.99, 1.31]). There was a main effect of phone ownership ($b = 0.53$ [0.19, 0.87], OR = 1.71 [1.21, 2.40]), indicating that youth who owned mobile phones at age 11–12 were 71 % more likely to report nicotine/tobacco use throughout the study period. While Fig. 1 suggests a growing difference between the groups, with nicotine/tobacco use more likely among phone owners, the interaction was not significant, perhaps in part due to low prevalence of reported use.

Next, we examined the association between mobile phone ownership at age 11–12 and cannabis use over time (Supplementary Table S). The phone*time interaction was not significant and was not retained. The resulting model indicated that cannabis use did not change over time ($b = 0.06$ [-0.15, 0.27], OR = 1.06 [0.86, 1.31]), and did not differ depending for youth who did and did not own mobile phones at age

11–12 ($b = 0.27$ [-0.20, 0.74], OR = 1.31 [0.82, 2.09]).

Social Media Use. We used a similar approach to evaluate the associations between age 11–12 social media use and substance use over time. Prevalence rates of substance use by social media status are shown in Fig. 2. In the model of the association between social media use and alcohol use over time (Table 3), we found a main effect of time ($b = -0.13$ [-0.20, -0.06], OR = 0.88 [0.82, 0.94]), a non-significant main effect of social media ($b = 0.04$ [-0.18, 0.26], OR = 1.06 [0.92, 1.21]), and a significant interaction between the two ($b = 0.09$ [0.01, 0.16], OR = 1.09 [1.01, 1.017]). This pattern indicates that there was a non-significant difference between the social media groups at age 11–12 that increased over time, consistent with Panel A of Fig. 2. Simple effects tests indicated a decline in alcohol use over time among both groups that was more pronounced among non-users of social media ($b = -0.14$ [-0.22, -0.07], OR = 0.82 [0.74, 0.90]) than among social media users ($b = -0.06$ [-0.13, -0.00], OR = 0.93 [0.87, 1.00]).

The model of the association between age 11–12 social media user status and nicotine/tobacco use is shown in Supplementary Table 3. This model produced a non-significant main effect of time ($b = 0.23$ [-0.09, 0.56], OR = 1.26 [0.91, 1.74]), a significant main effect of social media use ($b = 0.21$ [0.06, 0.37], OR = 1.22 [1.06, 1.37]), and a significant interaction between the two ($b = 0.08$ [0.05, 0.11], OR = 1.08 [1.05, 1.11]). Taken together, this suggests that social media users were more likely to report nicotine/tobacco use at age 11–12, and that this association grew stronger over time, consistent with Panel B in Fig. 2. Simple effects tests indicated that for the social media non-user group, there was no change in nicotine/tobacco use over time ($b = 0.25$ [-0.14, 0.64], OR = 1.08 [0.66, 1.70]). In contrast, for age 11–12 social media users the change was significant, with the likelihood of reporting nicotine/tobacco use increased by 19 % with each additional timepoint ($b = 0.17$ [0.02, 0.33], OR = 1.19 [1.02, 1.38]).

Finally, the model of the association between age 11–12 social media use and cannabis use over time (Supplementary Table 4) yielded a main effect of social media use status ($b = 1.59$ [1.04, 2.14], OR = 4.91 [2.84, 8.47]), but not of time ($b = 0.12$ [-0.10, 0.33], OR = 1.12 [0.91, 1.39]). The interaction between the two was not significant and thus not retained in the model. This pattern indicates that social media users were nearly five times more likely than non-users to report cannabis use at all four assessments from ages 11–12 to 13–14. Although Panel C of Fig. 2 suggests that this discrepancy grew over time the lack of interaction indicates the change was not statistically significant, perhaps at least partly because the prevalence of use was low.

Sensitivity Analyses. In the full sample, both predictors and substance use outcomes were first measured simultaneously at age 11–12, and evaluation of the potential impact of phone ownership and social media use at that age on subsequent substance use may be confounded by previous substance use. To address this concern, we conducted *post-hoc* sensitivity analyses in which we re-fit the models in a subsample comprised only of participants who reported at age 11–12 that they had never had more than a sip of alcohol, and never tried nicotine/tobacco or cannabis ($n = 9,628$, 87 % of the original sample).

First, we evaluated the association between phone ownership at age 11–12 and subsequent substance use. The phone ownership * time interaction did not predict alcohol use, indicating the association between phone ownership and alcohol use was consistent over time; the interaction term was not retained. There were main effects of both age 11–12 phone ownership ($b = 0.25$ [0.10, 0.41], OR = 1.28 [1.10, 1.50]) and time ($b = 0.59$ [0.53, 0.65], OR = 1.81 [1.70, 1.91]). The model of nicotine/tobacco use produced a similar pattern. The phone ownership * time interaction was not significant and not retained, but there were main effects of both age 11–12 phone ownership ($b = 0.65$ [0.22, 1.08], OR = 1.92 [1.25, 2.96]) and time ($b = 0.61$ [0.47, 0.76], OR = 1.84 [1.60, 2.13]). In contrast, the cannabis model yielded a main effect of time ($b = 0.87$ [0.64, 1.11], OR = 2.40 [1.89, 3.04]), but no main effect of age 11–12 phone ownership ($b = 0.54$ [-0.05, 1.13], OR = 1.71 [0.95, 3.09]) on cannabis use.

Table 1
Demographic and clinical characteristics at age 11–12.

Characteristic	M (SD) or Proportion
Sex assigned at birth	52.5 % male
Racial/ethnic background	53.5 % non-Hispanic white 19.7 % Hispanic or Latino 14.2 % Black or African American 10.5 % other racial/ethnic background
Household income	11.1 % less than \$25,000 36.3 % between \$25,000-\$99,999 31.1 % between \$100,000-\$200,000
Parent marital status	67.8 % married 13.9 % divorced or separated
Age at baseline	M = 12.0 years (SD = 0.67)
Grade at baseline (age 11–12)	13.7 % 5th grade 44.2 % 6th grade 35.4 % 7th grade
Age 11–12 phone ownership	71.2 %
Age 11–12 social media use	54.9 %
Any alcohol use	9.9 %
Any nicotine/tobacco use	0.9 %
Any cannabis use	0.5 %

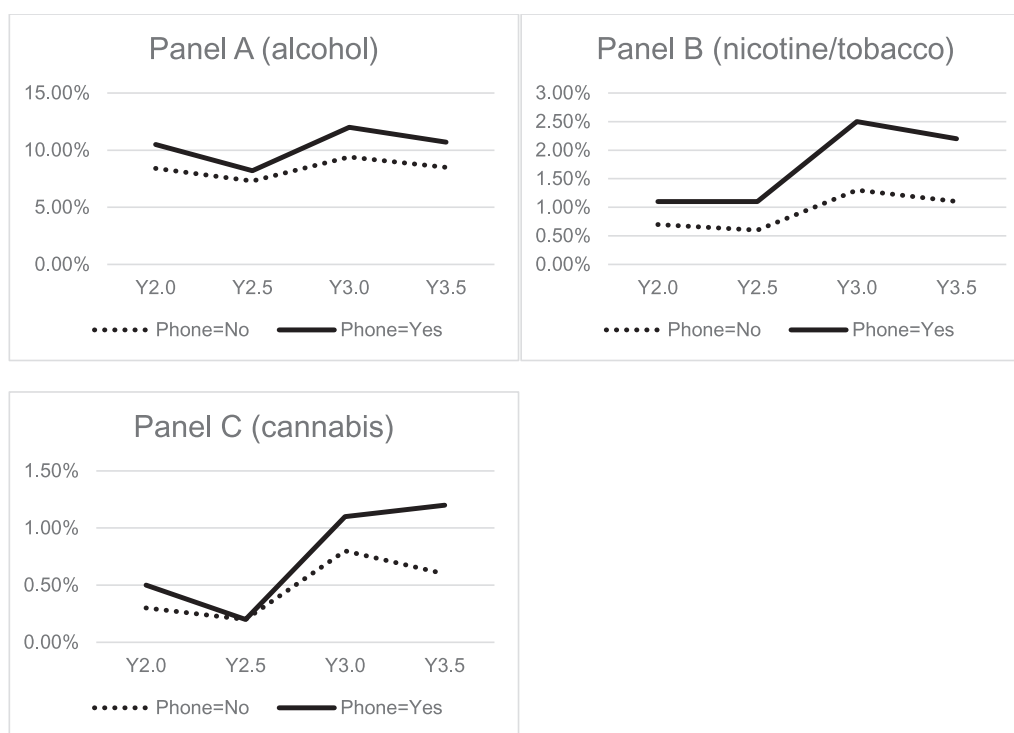


Fig. 1. Unadjusted prevalence of use of alcohol, nicotine/tobacco, and cannabis over time by phone ownership.

Table 2

GEE model of the association between mobile phone ownership and alcohol use over time.

Predictor	Odds Ratio (95 % c. i.)	Coefficient (95 % c. i.)	Std Err
Age	1.38 (1.27, 1.49)	0.32 (0.24, 0.40)	0.06
Biological sex	0.91 (0.82, 1.01)	-0.10 (-0.20, 0.01)	0.05
Household income	1.00 (1.00, 1.00)	0.00 (-0.01, 0.01)	<0.01
Racial/ethnic background			
Black or African American	0.24 (0.19, 0.31)	-1.41 (-1.66, -1.17)	0.03
Hispanic or Latino	0.53 (0.45, 0.62)	-0.64 (-0.79, -0.49)	0.04
Asian	0.37 (0.23, 0.60)	-0.99 (-1.48, -0.51)	0.09
Other	0.72 (0.61, 0.86)	-0.32 (-0.50, -0.15)	0.07
Time	0.91 (0.87, 0.96)	-0.09 (-0.14, -0.04)	0.02
Mobile phone ownership	1.34 (1.18, 1.51)	0.29 (0.16, 0.41)	

Note: c.i. = confidence interval; biological sex was coded as 0 = male, 1 = female; racial/ethnic background was coded as a categorical variable with non-Hispanic white as the reference category.

We then conducted a similar set of sensitivity analyses examining the association between social media use at age 11–12 and substance use over the next 18 months. In each case, the social media * time interaction was not significant and was not retained. We found main effects of both age 11–12 social media use ($b = 0.47$ [0.33, 0.61], $OR = 1.61$ [1.40, 1.85]) and time ($b = 0.59$ [0.53, 0.65], $OR = 1.81$ [1.71, 1.92]) on the likelihood of alcohol use. There were also main effects of both age 11–12 social media use ($b = 1.42$ [0.99, 1.84], $OR = 4.12$ [2.70, 6.27]) and time ($b = 0.62$ [0.47, 0.76], $OR = 1.85$ [1.60, 2.14]) on likelihood of nicotine/tobacco use. Similarly, there were significant main effects of both age 11–12 social media use [$b = 1.24$ [0.67, 1.81], $OR = 3.46$ [1.96, 6.10]] and time ($b = 0.88$ [0.64, 1.11], $OR = 2.40$ [1.90, 3.05]) on

the likelihood of cannabis use.

Overall, these analyses indicate that participants who at age 11–12 owned mobile phones but had never used any of the 3 substances were 28 % more likely to report alcohol use and 92 % more likely to report nicotine/tobacco use over the next 18 months compared with participants who did not own mobile phones at age 11–12. Similarly, participants who reported social media use and no prior substance use at age 11–12 were 61 % more likely to report alcohol use, 312 % more likely to report nicotine/tobacco use, and 246 % more likely to report cannabis use over the next 18 months relative to participants who reported no social media use at age 11–12.

4. Discussion

The goal of this secondary analysis was to conduct an initial evaluation of associations between mobile phone ownership and social media use at age 11–12, and alcohol, cannabis and nicotine and tobacco product use at that age and over the subsequent 18 months, among participants in the ABCD Study. We consistently found that youth who endorsed social media use at age 11–12, and to a lesser extent those who owned mobile phones, were more likely to report alcohol, nicotine/tobacco, and cannabis use over the study period. In some cases, this reflected associations that were stable over the full study period from age 11–12 to age 13–14 (e.g., associations between phone ownership and both alcohol and nicotine/tobacco use), and in other cases it reflected associations that grew stronger over time (e.g., associations between social media use and both alcohol and nicotine/tobacco use). Additionally, sensitivity analyses indicated that, among participants who had never used any of the 3 substances at age 11–12, mobile phone ownership, and particularly social media use, strongly predicted greater likelihood of use of these substances over the next 18 months.

These findings are consistent with previous studies focused on older youth and young adults, which also have reported positive associations between social media and digital device use and use of substances (Ranker et al., 2024; Willoughby et al., 2024). It is notable that these

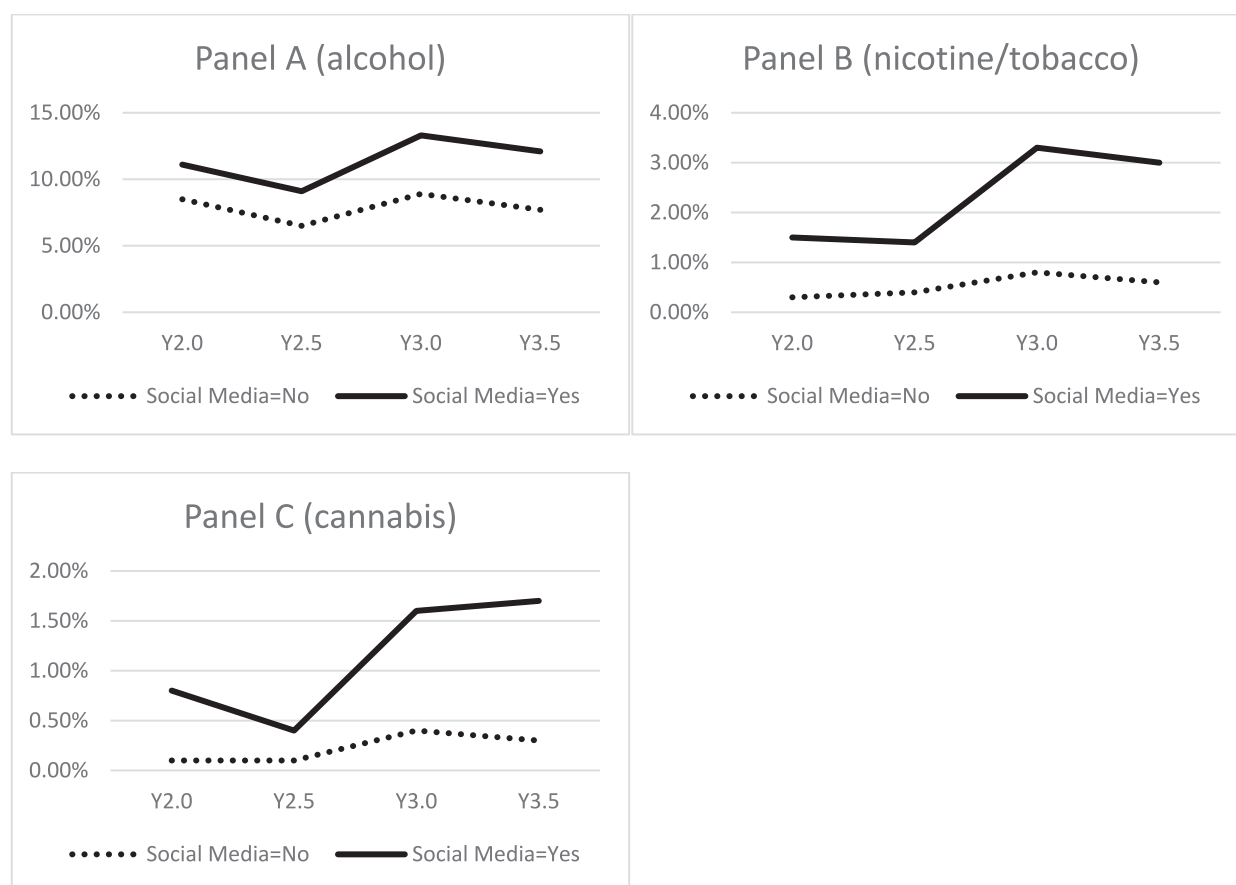


Fig. 2. Unadjusted prevalence of use of alcohol, nicotine/tobacco, and cannabis over time by social media user status.

Table 3

GEE model of the association between social media use and alcohol use over time.

Predictor	Odds Ratio (95 % c. i.)	Coefficient (95 % c. i.)	Std Err
Age	1.33 (1.23, 1.44)	0.28 (0.21, 0.36)	0.05
Biological sex	0.88 (0.79, 0.98)	-0.12 (-0.23, -0.02)	0.05
Household income	1.00 (1.00, 1.00)	0.00 (-0.01, 0.01)	<0.01
Racial/ethnic background			
Black or African American	0.22 (0.18, 0.29)	-1.49 (-1.74, -1.25)	0.03
Hispanic or Latino	0.50 (0.43, 0.58)	-0.69 (-0.84, -0.54)	0.04
Asian	0.38 (0.23, 0.62)	-0.97 (-1.46, -0.49)	0.09
Other	0.70 (0.58, 0.83)	-0.36 (-0.54, -0.18)	0.06
Time	0.88 (0.82, 0.94)	-0.13 (-0.20, -0.06)	0.03
Social media use	1.50 (1.31, 1.72)	0.41 (0.27, 0.54)	0.04
Social media use X time	1.09 (1.01, 1.17)	0.09 (0.01, 0.16)	0.04

Note: c.i. = confidence interval; biological sex was coded as 0 = male, 1 = female; racial/ethnic background was coded as a categorical variable with non-Hispanic white as the reference category.

associations were consistent in this younger sample despite low levels of use. The prevalence rates of all three substances were less than 10 % at age 11–12, with nicotine/tobacco and cannabis use reported by less than 1 % of the sample at that time. A substantial increase in substance use, across all substances, was observed over time among phone owners and social media users, with the most notable rise occurring during the

annual assessment at ages 12–13. This suggests that owning a mobile phone and using social media at age 11–12 may have conferred increased risk for subsequent substance use. Although it was not measured in this study, previous research suggests increased exposure to positive messages about use, and secondary impacts on perceived norms and acceptability, as a potential mechanism (Litt et al., 2023).

Social media use status was a more consistent predictor of substance use than phone ownership. This suggests that the ways in which a device is used may be more important compared with device ownership generally, and that ownership is less of a risk for substance use in the absence of social media. Previous studies have found that the relationship between phone ownership and distress may be mediated by social media use (Martin-Cardaba et al., 2024), suggesting phone ownership is less of a risk factor for those with limited social media use. A recent review suggests that the impact of social media consumption is also nuanced, and specific experiences can have positive or negative impact – for example, for youth with marginalized identities, social media use can provide more opportunities to find support and a sense of community (Choukas-Bradley et al., 2023). Some research has suggested that youth use of phones for purposes other than social media consumption (i.e., for communication with parents) may have more positive outcomes (Jensen et al., 2021).

If mobile phone ownership and social media use increase risk for use of these substances, their high prevalence across all youth highlight an urgent need for development of effective and flexible prevention and intervention programs with broad reach. Preliminary research suggests that programs, both online and offline, based in settings commonly accessed by youth (e.g., schools, primary care) may reduce substance use or mitigate known risk factors associated with it (Gilmore et al., 2023; Hawkins et al., 2015). Experts have also suggested legal and policy interventions that may be particularly relevant in terms of

reducing risks associated with social media (Costello et al., 2024). Additional research is needed to evaluate the impacts of such interventions on substance use outcomes.

It is important to note that the design of the present study does not permit us to conclude that mobile phone ownership or social media use at age 11–12 had a causal effect on subsequent increases in use of alcohol, nicotine/tobacco, or cannabis. An indirect causal impact of these factors is one possibility, where phone ownership and social media use serve as vehicles increasing exposure to positive messages about substance use that ultimately increase the odds of use. Other possibilities include that a third variable (e.g., parental oversight) may independently increase the likelihood of both the predictors and substance use outcomes. Additional research is needed to clarify the extent to which digital activity has a causal impact on youth substance use. Notably, the results of our post-hoc sensitivity analyses do suggest that mobile phone ownership and especially social media use among pre-teen youth are prospectively associated with subsequent use of alcohol, nicotine/tobacco, and cannabis.

Some aspects of the study limit generalizability of these findings. First, this was a secondary analysis of data drawn from a large cohort study, and as such the specific measures available to measure phone ownership and social media use were limited to assessments of any versus no use, potentially limiting our ability to detect differences related to type, frequency or intensity of use. Second, we relied on youth self-report measures of substance use; for nicotine/tobacco and cannabis in particular, low rates suggest the possibility of underreporting (Wade et al., 2023), though it is unclear that this would impact associations with phone and social media variables. Moreover, the reported rates are generally consistent with epidemiological data for this age range (Substance Abuse and Mental Health Services Administration, 2024). Notably, data collection occurred both before and during the first years of the COVID-19 pandemic, which has been associated with reduced substance use (Pelham et al., 2023); this may have diminished the prevalence of use in this sample. Third, we operationalized both mobile phone ownership and social media use as binary variables based on participants' behavior at age 11–12, based both on the available measures and on the idea that this age range reflects an important developmental period with the transition from elementary to middle school in the US. However, use of digital devices and social platforms can be nuanced, and may shift during transition from childhood to adolescence (Thomas et al., 2020). Our approach may have obscured nuanced relationships of frequency or content of digital behavior with substance use, or the potential impact of digital behavior that is more proximal to use. Relatedly, participants were younger than the average age of substance initiation (Rakerd-Richmond et al., 2017), and it is possible that these relationships may shift for older youth. Fourth, analyses did not account for some variables that may confound the hypothesized relationships. Examples include factors such as peer and parent substance use and participant mental health functioning, which were not consistently available across the study period. Follow-up studies should seek to account for the influence of these variables. Finally, different measures of substance use were employed at annual versus mid-year assessments. The lower prevalence observed at the first mid-year assessment might suggest that these measures underestimate actual substance use to a greater extent compared with the in-depth interview administered at the annual visits.

In this secondary analysis of data from the ABCD study, we found that owning a mobile phone and using social media at age 11–12 were each significant prospective predictors of substance use over the next 18 months. Social media use demonstrated a more consistent effect than did phone ownership and was a particularly strong predictor of increases in nicotine/tobacco and cannabis use. These data extend findings from earlier studies that focused on older adolescent and young adult samples and suggest that social media consumption and phone ownership among pre-teens may be associated with subsequent use of alcohol, nicotine/tobacco products, and cannabis. These findings highlight the importance

of additional research into digital and non-digital interventions to reduce these risks. The ubiquity of mobile phone ownership and social media use suggest that interventions with a broad reach (e.g., school-based) are likely to be most impactful.

CRediT authorship contribution statement

Neal Doran: Writing – review & editing, Writing – original draft, Project administration, Methodology, Funding acquisition, Formal analysis, Conceptualization. **Natasha E. Wade:** Writing – review & editing, Methodology, Data curation, Conceptualization. **Kelly E. Courtney:** Writing – review & editing, Conceptualization. **Ryan M. Sullivan:** Writing – review & editing, Software, Data curation. **Joanna Jacobus:** Writing – review & editing, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Neal Doran, Natasha Wade, Kelly Courtney, Joanna Jacobus reports financial support was provided by National Institutes of Health. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.addbeh.2024.108211>.

Data availability

Data will be made available on request.

References

- Alexander, J., Linkersdorfer, J., Toda-Thorne, K., Sullivan, R., Cummins, K., Tomko, R., Allen, N., Bagot, K., Baker, F., Fuemmeler, B., Hoffman, E., Kiss, O., Mason, M., Nguyen-Louie, T., Tapert, S., Smith, C., Squeglia, L., & Wade, N. (2024). Passively sensing smartphone use in teens with rates of use by sex and across operating systems. *Scientific Reports*, 14, 17982.
- Bagot, K., Tomko, R., Marshall, A., Hermann, J., Cummins, K., Ksinan, A., Kakalis, M., Breslin, F., Lisdahl, K., Mason, M., Redhead, J., Squeglia, L., Thompson, W., Wade, N., Tapert, S., Fuemmeler, B., & Baker, F. (2022). Youth screen use in the ABCD study. *Developmental Cognitive Neuroscience*, 57, Article 101150. <https://doi.org/10.1016/j.dcn.2022.101150>
- Bandura, A. (1977). *Social learning theory*. Prentice Hall.
- Cheng, B., Lim, C., Rutherford, B., Huang, S., Ashley, D., Johnson, B., Chung, J., Chan, G., Coates, J., Gullo, M., & Connor, J. (2024). A systematic review and meta-analysis of the relationship between youth drinking, self-posting of alcohol use and other social media engagement (2012–2021). *Addiction*, 119, 28–46. <https://doi.org/10.1111/add.16304>
- Choukas-Bradley, S., Kilic, Z., Stout, C., & Roberts, S. (2023). Perfect storms and double-edged swords: Recent advances in research on adolescent social media use and mental health. *Advances in Psychiatry and Behavioral Health*, 3, 149–157. <https://doi.org/10.1016/j.ypsc.2023.1003.1007>
- Costello, N., Sutton, R., Jones, M., Almossan, M., Raffoul, A., Ojumu, O., Salvia, M., Santoso, M., Kavanaugh, J., & Austin, S. (2024). Algorithms, addition and adolescent mental health: An interdisciplinary study to inform state-level policy action to protect youth from the dangers of social media. *American Journal of Law and Medicine*, 49(2–3), 135–172. <https://doi.org/10.1017/amj.2023.1025>
- Donaldson, C., Stuppelbein, D., Wilkinson, M., Zhang, Z., & Williams, R. (2023). Intersectional disparities in youth tobacco use by sexual and/or gender minority status and race and/or ethnicity. *Nicotine and Tobacco Research*, 25, 898–907. <https://doi.org/10.1093/ntr/ntac1264>
- Donaldson, S., Dormanesh, A., Perez, C., Majmundar, A., & Allem, J.-P. (2022). Association between exposure to tobacco content on social media and tobacco use: A systematic review and meta-analysis. *JAMA Pediatrics*, 176, 878–885. <https://doi.org/10.1001/jamapediatrics.2022.2223>
- Elmore, K., Scull, T., & Kupersmidt, J. (2017). Media as a “super peer”: How adolescents interpret media messages predicts their perception of alcohol and tobacco use norms. *Journal of Youth and Adolescence*, 46, 376–387. <https://doi.org/10.1007/s10964-10016-10609-10969>

- Garavan, H., Bartsch, H., Conway, K., Decastro, A., Goldstein, R., Heeringa, S., Jernigan, T., Potter, A., Thompson, W., & Zahs, D. (2018). Recruiting the ABCD sample: Design considerations and procedures. *Developmental Cognitive Neuroscience*, 32, 16–22. <https://doi.org/10.1016/j.dcn.2018.1004.1004>
- Gilmore, A., Fortson, K., Mullican, K., Garcia-Ramirez, G., Hutchins, A., Bartlett, A., Gooding, H., Wallis, E., Levy, S., Ruggiero, K., Kaysen, D., Danielson, C., Platner, R., Hartman, A., & Self-Brown, S. (2023). An ehealth prevention program for substance use, sexual assault, and sexual risk behaviors for adolescents in primary care: Pilot feasibility randomized controlled trial of teen well check. *JMIR Formative Research*, 7, e50833.
- Hanewinkel, R., Sargent, J., Hunt, K., Sweeting, H., Engels, R., Scholte, R., Mathis, F., Florek, E., & Morgenstern, M. (2014). Portrayal of alcohol consumption in movies and drinking initiation in low-risk adolescents. *Pediatrics*, 133, 973–982. <https://doi.org/10.1542/peds.2013-3880>
- Hawkins, J., Jensen, J., Catalano, R., Fraser, M., Botvin, G., Shapiro, V., Brown, C., Beardslee, W., Brent, D., Leslie, L., Rotheram-Borus, M., Shea, P., Shih, A., Anthony, E., Haggerty, K., Bender, K., Gorman-Smith, D., Casey, E., & Stone, S. (2015). *Unleashing the Power of Prevention. NAM Perspectives Discussion Paper*. National Academy of Medicine. <https://doi.org/10.31478/201506c>
- Henneberger, A., Mushonga, D., & Preston, A. (2020). Peer influence and adolescent substance use: A systematic review of dynamic social network research. *Adolescent Research Review*, 6, 57–73. <https://doi.org/10.1007/s40894-0019-00130-40890>
- Hu, Z., Jing, Y., Xue, Y., Fan, P., Wang, L., Vanyukov, M., Kirisci, L., Wang, J., Tarter, R. E., & Xie, X.-Q. (2020). Analysis of substance use and its outcomes by machine learning: II. Derivation and prediction of the trajectory of substance use severity. *Drug and Alcohol Dependence*, 206, Article 107604. <https://doi.org/10.1016/j.drugalcdep.2020.107604>
- Jackson, K., Janssen, T., & Gabrielli, J. (2018). Media/marketing influences on adolescent and young adult substance abuse. *Current Addiction Reports*, 5, 146–157. <https://doi.org/10.1007/s40429-0018-40199-40426>
- Jensen, M., George, M., Russell, M., Lippold, M., & Odgers, C. (2021). Daily parent-adolescent digital exchanges. *Research on Child and Adolescent Psychopathology*, 49, 1125–1138. <https://doi.org/10.1007/s10802-0020-00765-x>
- Kwan, I., Dickson, K., Richardson, M., MacDowall, W., Burchett, H., Stansfield, C., Brunton, G., Sutcliffe, K., & Thomas, J. (2020). Cyberbullying and children and young people's mental health: A systematic map of systematic reviews. *CyberPsychology Behavior & Social Networking*, 23. <https://doi.org/10.1089/cyber.2019.0370>
- Laestadius, L., Wahl, M., Pokhrel, P., & Cho, Y. (2019). From apple to werewolf: A content analysis of marketing for e-liquids on Instagram. *Addictive Behaviors*, 91, 119–127. <https://doi.org/10.1016/j.addbeh.2018.1009.1008>
- Lee, J., Krishnan-Sarin, S., & Kong, G. (2023). Social media use and cannabis vaping initiation among US youth. *Drug and Alcohol Dependence*, 249, Article 109949. <https://doi.org/10.1016/j.drugalcdep.2023.109949>
- Leonardi-Bee, J., Nderi, M., & Britton, J. (2016). Smoking in movies and smoking initiation in adolescents: Systematic review and meta-analysis. *Addiction*, 111, 1750–1763.
- Liang, K., & Zeger, S. (1986). Longitudinal data analysis using generalized linear models. *Biometrika*, 73, 13–22. <https://doi.org/10.1093/biomet/1073.1091.1013>
- Lisdahl, K., Tapert, S., Sher, K. J., Gonzalez, R., Nixon, S. J., Feldstein Ewing, S. W., Conway, K. P., Wallace, A., Sullivan, R., Hatcher, K., Kaiver, C., Thompson, W., Reuter, C., Bartsch, H., Wade, N. E., Jacobus, J., Albaugh, M. D., Allgaier, N., Anokhin, A. P., ... ABCD Consortium. (2021). Substance use patterns in 9-10 year olds: Baseline findings from the adolescent brain cognitive development (ABCD) study. *Drug and Alcohol Dependence*, 227, 108946. <https://doi.org/10.1016/j.drugalcdep.2021.108946>
- Litt, D., Zhou, Z., Fairlie, A., Waldron, K., Geusens, F., & Lewis, M. (2023). Experimental test of abstaining-and-drinking social media content on adolescent and young adult social norms and alcohol use. *Journal of Studies on Alcohol and Drugs*, 84, 700–709. <https://doi.org/10.15288/jsad.15223-00061>
- Martin-Cardaba, M., Martinez Dias, M., Perez, P., & Garcia Castro, J. (2024). Smartphone ownership, minors' well-being, and parental mediation strategies: An analysis in the context of social media influencers. *Journal of Youth and Adolescence*. <https://doi.org/10.1007/s10964-0024-02013-10967>
- McCabe, S. E., Schulenberg, J. E., Schepis, T. S., McCabe, V. V., & Veliz, P. T. (2022). Longitudinal analysis of substance use disorder symptom severity at age 18 years and substance use disorder in adulthood. *JAMA Network Open*, 5. <https://doi.org/10.22100/jamanetworkopen.222022.225324>
- Nasi, M., & Koivusilta, L. (2013). Internet and everyday life: The perceived implications of internet use on memory and ability to concentrate. *CyberPsychology Behavior & Social Networking*, 16, 88–93. <https://doi.org/10.1089/cyber.2012.0058>
- Pelham, W., Tapert, S., Zuniga, M., Thompson, W., Wade, N. E., Gonzalez, M. R., Patel, H., Baker, F., Dowling, G., Van Rinsveld, A., Baskin-Sommers, A., Kiss, O., & Brown, S. A. (2023). Pandemic-related changes in the prevalence of early adolescent alcohol and drug use, 2020–2021: Data from a multisite cohort study. *J Adolescent Health*, 73, 338–346. <https://doi.org/10.1016/j.jadohealth.2023.1002.1040>
- Pew Research Center (2021a). *Home broadband adoption, computer ownership vary by race, ethnicity in the U.S.* <https://www.pewresearch.org/short-reads/2021/07/16/home-broadband-adoption-computer-ownership-vary-by-race-ethnicity-in-the-u-s/>. Accessed August 5, 2024.
- Pew Research Center (2021b). *Lower-income Americans still lag in tech adoption.* <https://www.pewresearch.org/short-reads/2021/06/22/digital-divide-persists-even-as-americans-with-lower-incomes-make-gains-in-tech-adoption/>. Accessed August 2, 2024.
- Pew Research Center (2024). *Americans' social media use.* <https://www.pewresearch.org/internet/2024/01/31/americans-social-media-use/>. Accessed August 5, 2024.
- Purba, A., Thomson, R., Henery, P., Pearce, A., Henderson, M., & Katikireddi, S. (2023). Social media use and health risk behaviours in young people: Systematic review and meta-analysis. *BMJ*, 383, Doi: 10.1136/bmj-2022-073552.
- Rakerd-Richmond, L. S., Slutsky, W. S., & Wood, P. K. (2017). Age of initiation and substance use progression: A multivariate latent growth analysis. *Psychology of Addictive Behaviors*, 31, 664–675. <https://doi.org/10.1037/add0000304>
- Ranker, L., Wu, J., Hong, T., Wujaya, D., Benjamin, E., Bhatnagar, A., Robertson, R., Fetterman, J., & Xuan, Z. (2024). Social media use, brand engagement, and tobacco product initiation among youth: Evidence from a prospective cohort study. *Addictive Behaviors*, 154. <https://doi.org/10.1081016/j.addbeh.2024.1108000>
- Robinson, S., Sobell, L., Sobell, M., & Leo, G. (2014). Reliability of the Timeline Followback for cocaine, cannabis, and cigarette use. *Psychology of Addictive Behaviors*, 28, 154–162.
- Rutherford, B., Lim, C., Johnson, B., Cheng, B., Chung, J., Huang, S., Sun, T., Leung, J., Stjepanovic, D., & Chan, G. (2023). #TurnTrending: A systematic review of substance use portrayals on social media platforms. *Addiction*, 118, 206–217. <https://doi.org/10.1111/add.16020>
- Senekal, J., Groenewald, G., Wolfaardt, L., Jansen, C., & Williams, K. (2023). Social media and adolescent psychosocial development: A review. *South African Journal of Psychology*, 53, 157–171. <https://doi.org/10.1177/00812463221119302>
- Simons-Morton, B., & Chen, R. (2006). Over time relationships between early adolescent and peer substance use. *Addictive Behaviors*, 31, 1211–1223. <https://doi.org/10.1016/j.addbeh.2005.1209.1006>
- Sparrow, B., Liu, J., & Wegner, D. M. (2011). Google effects on memory: Cognitive consequences of having information at our fingertips. *Science*, 333, 776–778.
- Stevens, R., Brawner, B., Kranzler, E., Giorgi, S., Lazarus, E., Abera, M., Huang, S., & Ungar, L. (2020). Exploring substance use tweets of youth in the United States: Mixed methods study. *JMIR Public Health and Surveillance*, 6, e16191.
- Substance Abuse and Mental Health Services Administration (2024). *Key substance use and mental health indicators in the United States: Results from the 2023 National Survey on Drug Use and Health*.
- Sullivan, R., Wade, N., Wallace, A., Tapert, S., Pelham, W., Brown, S., Cloak, C., Feldstein Ewing, S., Madden, P., Martz, M., Ross, J., Kaiver, C., Wirtz, H., Heitzeg, M., & Lisdahl, K. (2022). Substance use patterns in 9 to 13-year-olds: Longitudinal findings from the Adolescent Brain Cognitive Development (ABCD) Study. *Drug and Alcohol Dependence Reports*, 5, Article 100120. <https://doi.org/10.1016/j.dadr.2022.100120>
- Thomas, G., Bennie, J., De Cocker, K., Ireland, M., & Biddle, S. (2020). Screen-based behaviors in Australian adolescents: Longitudinal trends from a 4-year follow-up study. *Preventive Medicine*, 141, Article 106258. <https://doi.org/10.1016/j.ypmed.2020.106258>
- Twenge, J., Joiner, T., Rogers, M., & Martin, G. (2018). Increases in depressive symptoms, suicide-related outcomes, and suicide rates among US adolescents after 2010 and links to increased new media screen time. *Clinical Psychological Science*, 6, 3–17. <https://doi.org/10.1177/2167702617723376>
- Valkenburg, P., Meier, A., & Beyens, I. (2022). Social media use and its impact on adolescent mental health: An umbrella review of the evidence. *Current Opinion in Psychology*, 44, 58–68. <https://doi.org/10.1016/j.copsych.2021.1008.1017>
- Vassey, J., Valente, T., Barker, J., Stanton, C., Li, D., Laestadius, L., Cruz, T., & Unger, J. (2023). E-cigarette brands and social media influencers on Instagram: A social network analysis. *Tobacco Control*, 32, e184–e191.
- Vogel, E., Barrington-Trimis, J., Vassey, J., Soto, D., & Unger, J. (2024). Young adults' exposure to and engagement with tobacco-related social media content and subsequent tobacco use. *Nicotine and Tobacco Research*, 26, S3–S12. <https://doi.org/10.1093/ntr/ntad1108>
- Wade, N., Sullivan, R., Tapert, S., Pelham, W., Huestis, M. A., Lisdahl, K., & Haist, F. (2023). Concordance between substance use self-report and hair analysis in community-based adolescents. *The American Journal of Drug and Alcohol Abuse*, 49, 76–84. <https://doi.org/10.1080/00952990.2023.202164931>
- Wang, Y., Duan, Z., Weaver, S., Self-Brown, S., Ashley, D., Emery, S., & Huang, J. (2022). Association of e-cigarette advertising, parental influence, and peer influence with US adolescent e-cigarette use. *JAMA Network Open*, 5, e2233938.
- Willoughby, J., Hust, S., Li, J., & Couto, L. (2024). Exposure to pro and anti-cannabis social media messages and teens' and college students' intentions to use cannabis. *Health Communication*, 39, 183–194. <https://doi.org/10.1080/10410236.10412022.12162707>
- Wilson, S., Malone, S. M., Venables, N. C., McGue, M., & Iacono, W. G. (2021). Multimodal indicators of risk for and consequences of substance use disorders: Executive functions and trait disinhibition assessed from preadolescence into early adulthood. *International Journal of Psychophysiology*, 163, 47–57. <https://doi.org/10.1016/j.ijpsycho.2019.1012.1007>